

14.1

Find Path and Sketch it.

Q1 $z(t) = (1 + \frac{1}{2}i)t$ $(2 \leq t \leq 5)$

Solution:

$$z(t) = u(t) + iy(t)$$

$$z(t) = (1 + \frac{1}{2}i)t$$

$$= \frac{t + \frac{t}{2}i}{2}$$

Here

$$u(t) = \frac{t}{2} \quad t = u$$

$$y(t) = \frac{t}{2} \quad y = \frac{u}{2}$$

→ Now we will see starting point $t=2$

$$u(2) = 2$$

$$y(2) = \frac{2}{2} = 1$$

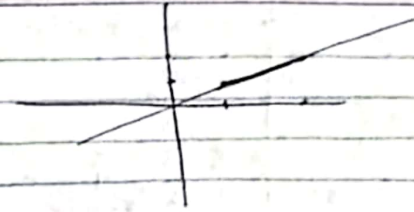
Starting point is $(2, 1)$

Now the ending point is for $t=5$

$$u(5) = 5$$

$$y(5) = \frac{5}{2} = 2.5$$

Ending point is $(5, 2.5)$



Q2 $z(t) = 3 + i + (1 - i)t$ $(0 \leq t \leq 3)$

Solution

$$z(t) = u(t) + iy(t)$$

$$z(t) = 3 + i + t - it$$

$$= 3 + t + i(1 - t)$$

Here

$$u(t) = 3 + t$$

$$y(t) = 1 - t$$

$$\therefore t = u - 3$$

put in $y(t) =$

$$y = 1 - (u - 3)$$

$$= 1 - u + 3$$

$$y = 4 - u$$

→ Now we will see starting point $t=0$

$$u(0) = 3$$

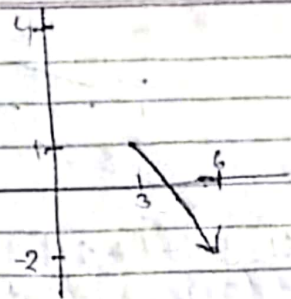
$$y(0) = 1$$

start point $(3, 1)$

→ Now ending point is $t=3$

$$u(3) = 3 + t = 3 + 3 = 6$$

$$y(3) = 1 - 3 = -2 \quad (6, -2)$$



Q3 $z(t) = t + 2it^2$ ($1 \leq t \leq 2$)

Solution:

$$z = x(t) + iy(t)$$

$$x(t) = t$$

$$y(t) = 2t^2$$

$$\therefore t = x$$

Put in $y(t)$

$$y = 2(x)^2$$

$$y = 2x^2$$

→ Now we will see starting point $t=1$

$$x(1) = 1$$

$$y(1) = 2(1)^2 = 2 \quad (1, 2)$$

→ Now will see ending point $t=2$

$$x(2) = 2$$

$$y(2) = 2(2)^2 = 8 \quad (2, 8)$$



Q4 $z(t) = t + (1-t)^2i$ ($-1 \leq t \leq 1$)

Solution:

$$z(t) = x(t) + iy(t)$$

$$x(t) = t$$

$$y(t) = (1-t)^2$$

$$\therefore t = x$$

Put in $y(t)$
 $y(t) = (1-x)^2$
 $t = -1$

Now we see starting point $t = -1$

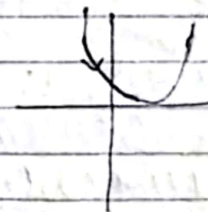
$$x(-1) = -1$$

$$y(-1) = (1+1)^2 = 4 \quad (-1, 4)$$

Now we see ending point $t=1$

$$x(1) = 1$$

$$y(1) = (1-1)^2 = 0 \quad (1, 0)$$



Q5 $z(t) = 3 - i + \sqrt{10}e^{-it}$ ($0 \leq t \leq 2\pi$)

Solution:

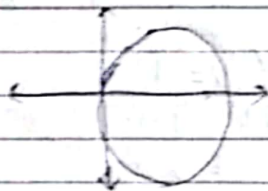
$$z(t) = x(t) + iy(t)$$

$$\begin{aligned} z &= 3 - i + \sqrt{10}e^{-it} \\ &= 3 - i + \sqrt{10}(\cos(-t) + i\sin(-t)) \\ &= 3 + \sqrt{10}\cos t + (-1 - \sqrt{10}\sin t)i \end{aligned}$$

$$\begin{aligned} x(t) &= 3 + \sqrt{10}\cos t \\ y(t) &= -1 - \sqrt{10}\sin t \end{aligned}$$

$$\Rightarrow (x-3)^2 + (y+1)^2 = 10$$

This is circle with centre $(3, -1)$, radius $\sqrt{10}$



Q6 $z(t) = 1 + i + e^{-\pi it}$ ($0 \leq t \leq 2$)

Solution

$$z(t) = x(t) + iy(t)$$

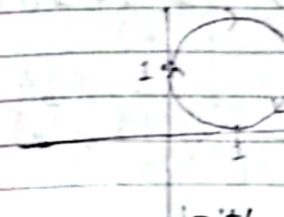
$$\begin{aligned} z(t) &= 1 + i + \cos(-\pi t) + i\sin(-\pi t) \\ &= [1 + \cos(\pi t)] + i[1 - \sin(\pi t)] \end{aligned}$$

$$\begin{aligned} x(t) &= 1 + \cos \pi t \\ y(t) &= 1 - \sin \pi t \end{aligned}$$

Now we see path on circle

$$(x-1)^2 + (y-1)^2 = 1$$

This is a circle with centre $(1, 1)$ with radius 1.



Q7 $z(t) = 2 + 4e^{i\pi t/2}$ ($0 \leq t \leq 2$)

Solution

$$z(t) = x(t) + iy(t)$$

$$\begin{aligned} z(t) &= 2 + 4 \left[\cos \frac{\pi t}{2} + i \sin \frac{\pi t}{2} \right] \\ &= 2 + 4 \cos \frac{\pi t}{2} + 4i \sin \frac{\pi t}{2} \\ &= \left(2 + 4 \cos \frac{\pi t}{2} \right) + i \left(4 \sin \frac{\pi t}{2} \right) \end{aligned}$$

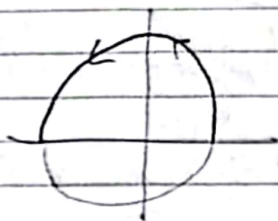
$$x(t) = 1 + 4 \cos \frac{\pi t}{2}$$

$$y(t) = 4 \sin \frac{\pi t}{2}$$

Now we see a path is on circle
 $(x-1)^2 + y^2 = 16$

→ This is a circle with centre $(1,0)$ & radius 4.

$$\rightarrow 0 \leq t \leq 2 \text{ so } 0 \leq \frac{\pi t}{2} \leq \pi$$



Q8 $z(t) = 5e^{-it}$ $0 \leq t \leq \pi/2$

Solution

$$z(t) = x(t) + iy(t)$$

$$\begin{aligned} z(t) &= 5e^{-it} \\ &= 5(\cos(-t) + i \sin(-t)) \\ &= 5 \cos t - 5i \sin t \end{aligned}$$

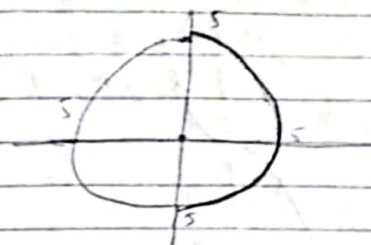
$$x(t) = 5 \cos t$$

$$y(t) = -5 \sin t$$

Now we will see that this path is on circle

$$x^2 + y^2 = 25$$

This is a circle with centre $(0,0)$ & radius 5.



Q9 $z(t) = t + it^3$ $(-2 \leq t \leq 2)$

Solution

$$z(t) = x(t) + iy(t)$$

$$z(t) = t + it^3$$

$$x(t) = t$$

$$y(t) = t^3$$

We see that this path is on graph

$$y = x^3$$

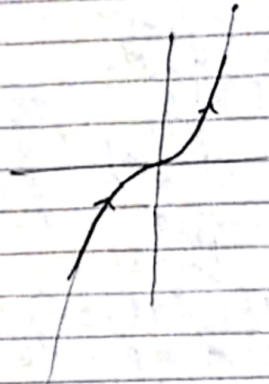
Now the starting point $t = -2$

$$x(-2) = -2$$

$$y(-2) = -8 \quad (-2, -8)$$

Now the ending point $t=2$

$$\begin{aligned}x(2) &= 2 \\y(2) &= 8\end{aligned} \quad (2, 8)$$



Q10 $z(t) = 2\cos t + i\sin t \quad (0 \leq t \leq 2\pi)$

Solution:

$$z(t) = x(t) + iy(t)$$

$$z(t) = 2\cos t + i\sin t$$

$$x(t) = 2\cos t$$

$$y(t) = \sin t$$

Now we can see this path is an ellipse

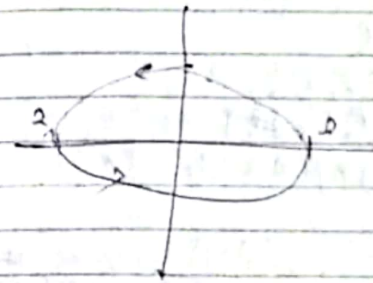
$$\frac{x^2}{4} + y^2 = 1$$

Starting point at $t=0$

$$\begin{aligned}x(0) &= 2 \\y(0) &= 0\end{aligned} \quad (2, 0)$$

Ending point at $t=2\pi$

$$\begin{aligned}x(2\pi) &= 2 \\y(2\pi) &= 0\end{aligned} \quad (2, 0)$$



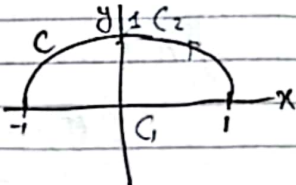
So $\ln(1-z)$ is analytic except on non positive number

but

$(1-z)$ is non-positive real number

it is analytic so Cauchy theorem applies and is 0.

Q22 $\oint \operatorname{Re} dz$, C :



Solution =

$$f(z) = \operatorname{Re} z$$

$$f(x+iy) = \operatorname{Re}(x+iy) = x$$

$$u(x,y) = x$$

$$v(x,y) = 0$$

$$\frac{\partial u}{\partial x} = 1$$

$$\frac{\partial v}{\partial y} = 0$$

$$1 \neq 0$$

So it is not analytic

Now

$$z(t) = \cos t + i \sin t$$

$$z'(t) = -\sin t + i \cos t$$

$(-1,0)$ to $(1,0)$

$$z_1(t) = t$$

$$z_1'(t) = 1$$

Now putting values

$$\oint f(z) = \int_0^\pi f(z_1(t)) z_1'(t) dt + \int_{-1}^1 f(z_2(t)) z_2'(t) dt$$

$$= \int_0^\pi \operatorname{Re}(\cos t + i \sin t) (-\sin t + i \cos t) dt + \int_{-1}^1 \operatorname{Re}(t) \cdot 1$$

$$= \int_0^\pi (-\sin t \cos t + i \cos^2 t) dt + \int_{-1}^1 t dt$$

$$= \int_0^\pi \sin t \cos t dt + i \int_0^\pi \cos^2 t dt + \int_{-1}^1 t dt$$

$$\textcircled{1} = \frac{1}{2} \int_0^\pi \sin 2t dt = \frac{1}{4} \cos 2t \Big|_0^\pi = \frac{1}{4} (\cos 2\pi - \cos 0) = 0$$

$$\textcircled{2} = \int_0^\pi \cos^2 t dt + 1$$

$$\cos^2 t = \frac{1 + \cos 2t}{2} = \frac{1}{2} + \frac{1}{2} \cos 2t$$

$$\int_0^\pi \cos^2 t dt = \frac{1}{2} \int_0^\pi (1 + \cos 2t) dt = \frac{1}{2} \left(t + \frac{\sin 2t}{2} \right) \Big|_0^\pi = \frac{1}{2} (\pi + 0) = \frac{\pi}{2}$$

Q30 $\int \frac{2z^3 + z^2 + 4}{z^4 + 4z^2} dz$, $C: |z|=4$ CC
Use partial fraction

Solution:

$$= \frac{1}{4} \int \frac{2z^3 + z^2 + 4}{z^2} dz - \frac{1}{4} \int \frac{2z^3 + z^2 + 4}{z^2 + 4} dz$$

① ②

Now

$$= \frac{1}{4} \int \frac{2z^3 + z^2 + 4}{z^2} dz$$

if $z = e^{i\theta}$, $dz = ie^{i\theta} d\theta$, $\frac{dz}{i} = e^{i\theta} d\theta$

$$\int \frac{2z^3 + z^2 + 4}{z^2} dz = \int 2z dz + \int \frac{z^2}{z^2} dz + \int \frac{4}{z^2} dz$$

$$= 2 \int z dz + \int dz + 4 \int z^{-2} dz$$

$$= 2 \int_0^{2\pi} e^{i\theta} d\theta$$

$$= \frac{2e^{2i\theta}}{i} \Big|_0^{2\pi} + \frac{e^{i\theta}}{i} \Big|_0^{2\pi} + \frac{4e^{-i\theta}}{-i} \Big|_0^{2\pi}$$

$$= \frac{2e^{4i\pi} - 1}{i} + \frac{e^{2i\pi} - 1}{i} + \frac{4e^{-2i\pi} - 1}{-i}$$

$$= 0$$

② by putting values in ②

$$= \frac{1}{4} (0)$$

$$= 0$$

Q3 $|z+1| = 1.4$

Sol

$$\int \frac{z^2}{(z+1)(z-1)} dz$$

$$\frac{z^2}{(z+1)(z-1)} = \frac{A}{(z+1)} + \frac{B}{z-1} \quad \text{--- (1)}$$

$$z^2 = A(z-1) + B(z+1)$$

$$1 = A(-2)$$

$$\boxed{A = -\frac{1}{2}}$$

$$B = \frac{1}{2}$$

Put in (1)

$$= -\frac{1}{2} \int \frac{1}{z+1} dz + \frac{1}{2} \int \frac{1}{z-1} dz$$

$$= \int_0^1 \frac{-\frac{1}{2}}{z-(-1)} dz + \int_0^1 \frac{\frac{1}{2}}{z-1} dz$$

$$= 2\pi i f(-1) = 2\pi i f(1)$$

$$= 2\pi i \times -\frac{1}{2} = 2\pi i \times \frac{1}{2}$$

$$= -\pi i = \pi i$$

$$= \pi i - \pi i$$

$$\boxed{V = 0}$$

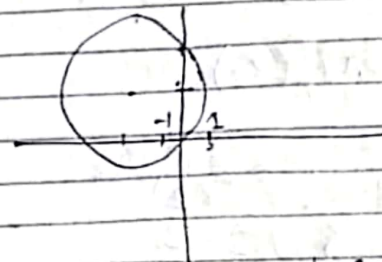
Q4 $|z+5-5i| = 7$

$$z^2 - 1 = (z-1)(z+1)$$

Solution

$5+5i$ are centre and 7 is radius

and also plot 1 and -1



So -1 lies inside circle we use $(z+1)$

$$\int \frac{z^2}{z^2-1} dz = \int \frac{f(z)}{z+1} dz = 2\pi i f(-1) \quad \text{--- (1)}$$

$$\text{Now } f(-1) = \frac{(-1)^2}{-1-1} = \frac{(-1)^2}{-2} = \boxed{-\frac{1}{2}}$$

put in (1)

$$= 2\pi i \left(-\frac{1}{2}\right)$$

$$\boxed{= -\pi i}$$

14.3

Contour Integration

Q1 Integrate $\frac{z^2}{(z^2-1)}$ by Cauchy's formula
CCV around circle.

$$Q1 \quad |z+1| = 1 \quad \oint \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0)$$

Solution $|z+1| = 1$

$$\oint \frac{z^2}{z^2-1} dz = \oint \frac{z^2}{(z+1)(z-1)} dz$$

Here $f(z) = z^2$

$$z_0 = 2$$

$$z_0 = -1$$

Now

$$\oint \frac{f(z)}{z+1} dz = 2\pi i f(-1) \quad \text{--- (1)}$$

But

$$f(-1) = \frac{(-1)^2}{(-1)^2 - 1} = \frac{f(z)}{z-z_0} = \frac{z^2}{z-z_0}$$

$$= \frac{-1}{2}$$

put in (1)

$$\oint \frac{f(z)}{(z+1)^2} dz = 2\pi i \cdot -\frac{1}{2}$$

$$\boxed{= -\pi i}$$

$$Q \quad |z-1-i| = \pi/2$$

Solution

$$\frac{z^2}{z^2-1} = \frac{z^2}{(z-1)(z+1)} \quad f(z) = \frac{z^2}{z+1}$$

$$|z-1| = 2$$

Here $2\pi i f(z_0) = \oint \frac{f(z)}{z-z_0}$ general

$$2\pi i f(1) = \oint \frac{f(z)}{z-1} \quad \text{--- (1)}$$

Now

$$f(1) = \frac{z^2}{z^2-1} = \frac{1}{1-1} = \int \frac{z^2}{(z-1)(z+1)} dz$$

$$= \frac{z^2}{z+1}$$

$$= \frac{1}{1+1}$$

$$f(1) = \frac{1}{2}$$

Putting values in (1)

$$= 2\pi i \times \frac{1}{2}$$

$$\boxed{= \pi i}$$

$$= \frac{1}{2} 2\pi i f(z_0)$$

$$= \pi i \left(\frac{1}{2}\right)^3$$

$$= \pi i \frac{1}{8}$$

$$= \frac{\pi}{8} i$$

$$\boxed{I = \frac{\pi}{8}}$$

Q 8 $(z^2 \sin z)/4z-1$

Solution =

$$\int \frac{z^2 \sin z}{4z-1} dz = \int \frac{z^2 \sin z}{4(z-\frac{1}{4})}$$

$$\int \frac{f(z)}{z-z_0} = 2\pi i f(z_0)$$

$$\frac{1}{4} \int \frac{z^2 \sin z}{z-\frac{1}{4}} = \frac{1}{4} \int \frac{f(z)}{z-z_0}$$

$$= \frac{1}{4} 2\pi i f(z_0)$$

$$= \frac{\pi i}{32} \sin \frac{1}{4}$$

$$\boxed{I = 0.024i}$$

Q 11 $\int \frac{dz}{z^2+4}$, $C: 4u^2+(y-2)^2=4$

Solution

$$\frac{z^2 = -4}{z = \pm 2i}$$



Here $C: 4u^2+(y-2)^2=4$, where only $z=+2i$ lies inside

$$\int \frac{dz}{(z^2+4)(z-2i)} = \int \frac{dz}{z-2i}$$

$$\Rightarrow \int \frac{f(z)}{z-z_0} = 2\pi i f(z_0)$$

$$\int \frac{dz}{z-2i} = 2\pi i f(z_0)$$

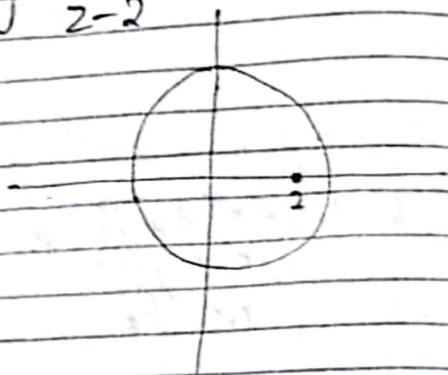
$$= 2\pi i \left[\frac{1}{z+2i} \right]$$

$$= 2\pi i \left[-\frac{1}{4} i \right]$$

$$= \frac{\pi}{2}$$

Q13 $\int \frac{z+2}{z-2} dz$ $C: |z-1|=2$

Solution



$$\int \frac{z+2}{z-2} dz$$

$$\oint \frac{f(z)}{z-z_0} = 2\pi i f(z_0)$$

$$\begin{aligned} \int \frac{z+2}{z-2} &= \oint \frac{f(z)}{z-z_0} \\ &= 2\pi i f(z_0) \\ &= 2\pi i [z+2] \\ &= 2\pi i [2+2] \\ &= 2\pi i [4] \\ &= 8\pi i \end{aligned}$$

Q14 $\int \frac{e^z}{ze^z-2iz} dz$ $C: |z|=0.6$

Sol

$$\int \frac{e^z}{ze^z-2iz} = \int \frac{e^z}{(e^z-2i)(z-0)} \quad f(z) = \frac{e^z}{e^z-2i}$$

$$\oint \frac{f(z)}{z-z_0} = 2\pi i f(z_0)$$

$$\int \frac{e^z}{ze^z-2iz} = \oint \frac{f(z)}{z-z_0}$$

$$= 2\pi i f(z_0)$$

$$= 2\pi i \left(\frac{1}{1-2i} \right)$$

$$= \frac{2\pi i (1+2i)}{5}$$

$$= \frac{-4\pi + 2i\pi}{5}$$

$$\boxed{= \frac{-4\pi + 2i\pi}{5}}$$

Q15 $\oint \frac{\cosh(z^2 - \pi i)}{z - \pi i} dz$, C in boundary

of square vertices $\pm 2, \pm 2i$

Sol

$$\oint \frac{\cosh(z^2 - \pi i)}{z - \pi i} dz \quad f(z) = \cosh(z^2)$$

$$\int \frac{f(z)}{z - z_0} = 2\pi i f(z_0)$$

$$\begin{aligned} \oint \frac{\cosh(z^2 - \pi i)}{z - \pi i} dz &= 2\pi i f(\pi i) \\ &= 2\pi i \cosh((\pi i)^2 - \pi i) \\ &= 2\pi i \cosh(-\pi - \pi i) \end{aligned}$$

Inte Around unit circle
Q5 $(\cos 3z)/6z$

Solution

$$\int \frac{\cos 3z}{6z} dz = \frac{1}{6} \int \frac{\cos 3z}{z-0} dz$$

$$\int \frac{f(z)}{z-z_0} = 2\pi i f(z_0)$$

$$\frac{1}{6} \int \frac{\cos 3z}{z-0} = \frac{1}{6} \int \frac{f(z)}{z-z_0}$$

$$f(z) = \cos 3z \quad z_0 = 0$$

Now

$$= \frac{1}{6} \int \frac{f(z)}{z-z_0}$$

$$= \frac{1}{6} 2\pi i f(z_0)$$

$$= \frac{\pi \cdot i \cdot (1)}{3}$$

$$= \frac{\pi i}{3}$$

Q6 $\frac{e^{2z}}{(\pi z - i)}$

Solution

$$\int \frac{e^{2z}}{\pi z - i} dz = \frac{1}{\pi} \int \frac{e^{2z}}{z - i/\pi} dz$$

$$\int \frac{f(z)}{z-z_0} = 2\pi i f(z_0)$$

$$\frac{1}{\pi} \int \frac{e^{2z}}{z - i/\pi} dz = \frac{1}{\pi} \int \frac{f(z)}{z-z_0}$$

$$= \frac{1}{\pi} 2\pi i f(z_0)$$

$$= 2i e^{2i/\pi}$$

$$= 2i (\cos(2/\pi) + i \sin(2/\pi))$$

$$= -1.18 + 1.6i$$

Q#7 $\frac{z^3}{(2z - i)}$

Solution

$$\int \frac{z^3}{2z - i} dz = \frac{1}{2} \int \frac{z^3}{z - i/2} dz$$

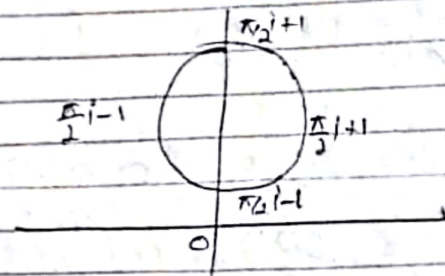
$$\int \frac{f(z)}{z-z_0} = 2\pi i f(z_0)$$

$$\frac{1}{2} \int \frac{z^3}{z - i/2} = \frac{1}{2} \int \frac{f(z)}{z-z_0}$$

Q26 $\oint \coth \frac{1}{2} z \, dz$, C the circle $|z - \frac{1}{2}\pi i| = \frac{1}{2}$ clockwise.

Solution

$\coth 0.5z = \frac{\cosh 0.5z}{\sinh 0.5z}$ is not analytic $z=0$ that lies outside circle $|z - \frac{1}{2}\pi i| = \frac{1}{2}$



So $\oint \coth 0.5z \, dz = 0$

Q27 $\oint \frac{\cos z}{z} \, dz$, C consist of $|z|=1$ CCW and $|z|=3$ CV

Solution:

$$\oint_{C_1} \frac{\cos z}{z} \, dz = \oint_{C_2} \frac{\cos z}{z} \, dz$$

$$\oint \frac{\cos z}{z} \, dz = \oint_{C_1} \frac{\cos z}{z} \, dz + \oint_{C_2} \frac{\cos z}{z} \, dz = \oint_{C_1} \frac{\cos z}{z} \, dz - \oint_{C_2} \frac{\cos z}{z} \, dz = 0$$

$$\text{(because } \oint_{C_2} \frac{\cos z}{z} \, dz = - \oint_{C_1} \frac{\cos z}{z} \, dz \text{)}$$

$$\boxed{\oint \frac{\cos z}{z} \, dz = 0}$$

Q28 $\oint \frac{\tan^2 z}{z^4 - 16} \, dz$, C the boundary

of square with vertices $\pm 1, \pm i$ CV.

$$z^4 - 16 = 0$$

$$z^4 = 16$$

$$\boxed{z^2 = \pm 4}$$

$$z^2 = 4$$

$$\boxed{z = \pm 2}$$

$$z^2 = -4$$

$$\boxed{z = \pm 2i}$$

As we see vertices in Q and see radius we can say points outside square so Cauchy theorem

is analytic $\oint \frac{\tan^2 z}{z^4 - 16} \, dz = 0$

Q24 $\int \frac{dz}{z^2-1}$, $C:$

$$\frac{1}{z^2-1} = \frac{1}{(z-1)(z+1)}$$

$$= \frac{A}{z-1} + \frac{B}{z+1}$$

$$1 = A(z+1) + B(z-1)$$

$$\begin{aligned} A+B &= 0 \Rightarrow A = -B \\ A-B &= 1 \\ -B-B &= 1 \\ -2B &= 1 \\ B &= -\frac{1}{2} \\ A &= \frac{1}{2} \end{aligned}$$

Putting values

$$= \frac{1}{2(z-1)} - \frac{1}{2(z+1)}$$

Now

$$\int_C \frac{dz}{z^2-1} = \int_{C_1} \frac{dz}{z^2-1} - \int_{C_2} \frac{dz}{z^2-1}$$

For C_1 :

$$\int \frac{dz}{z^2-1} = 2\pi i$$

Now C_2

$$\int \frac{dz}{z^2-1} = -2\pi i$$

Now

$$\begin{aligned} &= C_1 - C_2 \\ &= (2\pi i - (-2\pi i)) / 2 \\ &= \frac{4\pi i}{2} \\ &= 2\pi i \end{aligned}$$

Q25 $\int \frac{e^z}{z} dz$, C consist of $|z|=2$ CCW and $z=1$ clockwise

Solution:

$$\int_C \frac{e^z}{z} dz = \int_{C_1} \frac{e^z}{z} dz - \int_{C_2} \frac{e^z}{z} dz$$

and e^z is analytic everywhere so $\frac{e^z}{z}$ is analytic

$$\text{So } \oint_C \frac{e^z}{z} dz = 0$$

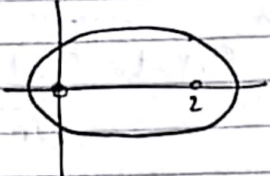
we cannot apply Cauchy integral theorem.

$$\begin{aligned}\int_0^{\pi} \cos^2 t dt &= \frac{1}{2} \int_0^{\pi} dt + \frac{1}{2} \int_0^{\pi} \cos 2t dt \\&= \frac{1}{2} t \Big|_0^{\pi} + \frac{1}{4} \sin 2t \Big|_0^{\pi} \\&= \frac{1}{2} (\pi - 0) + \frac{1}{4} (\sin 2\pi - \sin 0) \\&= \frac{\pi}{2}\end{aligned}$$

$$\textcircled{3} \int_1^{-1} t dt = \left[\frac{t^2}{2} \right]_1^{-1} = \left(\frac{1^2}{2} - \frac{(-1)^2}{2} \right) = 0$$

$$\left| \int_C f(z) dz = \frac{\pi i}{2} \right|$$

Q 23 $\int \frac{2z-1}{z^2-z} dz$ C:



Sol

$$\frac{2z-1}{z(z-1)} = \frac{A}{z} + \frac{B}{z-1}$$

$$(2z-1) = A(z-1) + Bz$$

$$2z-1 = (A+B)z - A$$

$$-A = -1$$

$$\boxed{A=1}$$

$$A+B=2$$

$$B+1=2$$

$$B=2-1$$

$$\boxed{B=1}$$

Now putting values

$$\int \frac{2z-1}{z^2-z} dz = \int \frac{dz}{z} + \int \frac{dz}{z-1} \quad \text{--- (1)}$$

$$\textcircled{1} \oint \frac{dz}{z} = \oint z^{-1} dz = 2\pi i$$

$$\textcircled{2} \oint \frac{dz}{z-1} = \oint (z-1)^{-1} dz = 2\pi i$$

Now put in (1)

$$\begin{aligned}&= 2\pi i + 2\pi i \\&\boxed{= 4\pi i}\end{aligned}$$

Q21 $\oint \frac{dz}{2-3i}$, C the circle $|z|=\pi$ counter-clockwise

Solution:

$$|z| = \pi$$

$$|x+iy| = \pi$$

$$\sqrt{x^2+y^2} = \pi$$

$$x^2+y^2 = \pi^2$$

$$R = \pi$$

The function

$$f(z) = \frac{1}{2-3i}$$

$$2-3i \neq 0$$

$$2 \neq 3i$$

which is $3 < \pi \rightarrow$

So lies in given circle

Q20 $\oint \ln(1-z) dz$ the boundary of parallelogram

$1-z$ is entire and is therefore analytic everywhere

$\ln z$ is analytic except on set non-positive numbers

Now for unit circle

$$z(t) = e^{it} \Rightarrow \text{img}(z) = \sin t$$

$$z'(t) = ie^{it}$$

Now

$$\int_0^{2\pi} \sin t - 1(\cos t + i \sin t) dt$$

$$= \int_0^{2\pi} i \sin t d(\sin t) + \int_0^{2\pi} i^2 \sin^2 t dt$$

$$= \int_0^{2\pi} \sin t d(\sin t) + i^2 \int_0^{2\pi} \sin^2 t dt$$

$$= -\frac{1}{2} 2\pi$$

$$\boxed{f = -\pi}$$

Q16 $f(z) = \frac{1}{(\pi z - 1)}$

Solution

$$\int f(z) = \frac{1}{\pi} \int \frac{1}{z - 1/\pi} dz$$

$$= \frac{1}{\pi} (2\pi i)$$

$$\boxed{f = 2i}$$

Q17 $f(z) = \frac{1}{|z|^2}$

Same as Q14

Q18 $f(z) = \frac{1}{(4z^2 - 3)}$

Sol

$$\int f(z) = \int \frac{1}{4z^2 - 3}$$

$$= \frac{1}{4} \int \left(\frac{1}{z^2 - 3/4} \right) dz$$

$$= \frac{1}{4} (2\pi i)$$

$$\boxed{= \frac{\pi i}{2}}$$

Q19 $f(z) = z^3 \cot z$

must be ho.

$$= i \int_0^{2\pi} e^{-3it} \cdot e^{it} dt$$

$$= i \int_0^{2\pi} e^{-2it} dt$$

$$= i \cdot \frac{e^{-2it}}{-2i} \Big|_0^{2\pi}$$

$$= i \left(\frac{e^{-4\pi i}}{-2i} - \frac{e^0}{-2i} \right)$$

$$= i \left(\frac{1}{-2i} - \frac{1}{-2i} \right)$$

$$\boxed{= 0}$$

Q13 $f(z) = 1/(z^4 - 1.1)$

Solution

$$z^4 - 1.1 = 0$$

$$z^4 = 1.1$$

$$1.1 = |1.1|$$

$$|z^4| = |z|^4$$

$$\boxed{|z| = \sqrt[4]{1.1} > 1}$$

So we cannot apply theorem

Q14 $f(z) = \frac{1}{z}$

Solution

It is not even defined in 0, so it is not analytic

So Cauchy's theorem is not applied

$$z = e^{it} \quad z'(t) = ie^{it}$$

Now putting values

$$\boxed{\int f(z) dz = 0}$$

Q15 $f(z) = \text{Im}(z)$

$$f(z) = f(x+iy)$$

$$= \text{Im}(x+iy)$$

$$= y$$

$$\frac{\partial u}{\partial x} = 0$$

$$\frac{\partial u}{\partial y} = 1$$

$$0 \neq 1$$

not analytic Cauchy's theorem not applied

14.2

Cauchy Theorem applicable?

~~Q10: $f(z) = \exp(-z^2)$~~

Q11: $f(z) = \frac{1}{(z-1)^2}$

$$\int \frac{1}{(z-1)^2} dz = \frac{1}{2} \int \frac{1}{z-1} dz$$

$z = \frac{1}{2}$ is a singularity that lies inside unit circle ($|z|=1$)

So

$$= \frac{1}{2} \int \frac{1}{z-1/2} dz$$

$$= \frac{1}{2} (2\pi i)$$

$$= \pi i$$

Q12: $f(z) = z^{-3}$

Solution:

$$f(z) = f(x+iy)$$

$$= \overline{(x+iy)^3}$$

$$= (x-iy)^3$$

$$= x^3 - 3x^2yi + 3xy^2i^2 - y^3i^3$$

$$= x^3 - 3x^2yi - 3xy^2 + y^3i$$

$$= (x^3 - 3xy^2) + (y^3 - 3x^2y)i$$

$$U(x,y) = x^3 - 3xy^2$$

$$V(x,y) = y^3 - 3x^2y$$

$$\frac{\partial U}{\partial x} = 3x - 3y^2$$

$$\frac{\partial V}{\partial y} = 3y^2 - 3x^2$$

$$\frac{\partial U}{\partial y} = 6xy$$

$$\frac{\partial V}{\partial x} = -6xy$$

This is not analytic

→ representation of unit circle in counterclockwise:

$$z(t) = e^{it} \quad t \in [0, 2\pi]$$

Also

~~dot z~~ $z'(t) = ie^{it}$

Now putting values

$$\int_0^{2\pi} (e^{it})^3 ie^{it} dt$$

$$\int_0^{2\pi} (e^{-it})^3 ie^{it} dt$$

$$= \frac{1}{2} \exp(w) \Big|_1^4$$

$$= \frac{1}{2} \exp(4) - \exp(1) =$$

$$= \sinh 1$$

Q24 $\int \cos z dz$, C is semicircle $|z| = \pi$
u do from
- π to π

Solution

$$\frac{d}{dz} \left(\frac{1}{2} \sin 2z \right) = \cos 2z$$

$$F(z) = \frac{1}{2} \ln z \quad f(z) = \cos z$$

→ From (1)

$$F'(z) = \cos z = f(12)$$

Now taking derivative

$$\int_0^{\pi} \cos 2z \, dz = \frac{1}{2} \sinh^2(\pi) - \frac{1}{2} \sinh^2(-\pi) = \sinh^2 \pi$$

$$= e^{i\pi i} - e^{-i\pi i} \quad (2)$$

$$= \frac{e^{2i} - e^{0i}}{2i} \cdot i$$

$$= \frac{e^{-2i} - 2e^{2i}}{2i^2} \cdot i$$

$$= \frac{e^{-2\pi} - e^{2\pi}}{2} \cdot i = \frac{e^{2\pi} - e^{-2\pi}}{2} \cdot i$$

$$\boxed{= i \sinh 2\pi}$$

Q23: $\int e^z dz$, C the shortest path from πi to $2\pi i$

Solution

Solution $f(z) = e^z$ & $F(z) = e^z$

$$F'(z) = e^z = f(z)$$

Now integrate

$$\begin{aligned}\int_C e^z &= F(2\pi i) - F(0) \\ &= e^{2\pi i} - e^{0i} \\ &= \cos 2\pi + i \sin 2\pi - (\cos 0 + i \sin 0) \\ &= 1 + i \cdot 0 - (1 + i \cdot 0) \\ &= 0\end{aligned}$$

$$z'(t) = -ie^{-it}$$

$$f(z) = \int_0^{2\pi} f(z(t)) z'(t) dt$$

$$= \int_0^{2\pi} f(e^{-it}) (-ie^{-it}) dt$$

$$= -i \int_0^{2\pi} (e^{-it} + (e^{-it})^2) e^{-it} dt$$

$$= -i \int_0^{2\pi} (e^{-2it} + e^{-it}) dt$$

$$= -i \int_0^{2\pi} (e^{-2it} + 1) dt$$

$$= -i \int_0^{2\pi} e^{-2it} dt - i \int_0^{2\pi} 1 dt$$

$$= -i \left[\frac{e^{-2it}}{-2i} \right]_0^{2\pi} - i \left[t \right]_0^{2\pi}$$

$$= \frac{e^{-2i \cdot 2\pi}}{2} - \frac{e^{-2i \cdot 0}}{2} - i(2\pi - 0)$$

$$= \frac{e^{-4i\pi}}{2} - \frac{1}{2} - 2\pi i$$

$$= \frac{1}{2} - \frac{1}{2} - 2\pi i$$

$$\boxed{-2\pi i}$$

Q25 $\int_C z \exp(z^2) dz$, C from 1 along the axis to i

Solution:

$$\begin{aligned} f(z) &= z \exp(z^2) = (x+iy) \exp(x^2-y^2+2xyi) \\ &= (x+iy) \exp(x^2-y^2) (\cos(2xy) + i \sin(2xy)) \\ &= (x+iy) \exp(x^2-y^2) (\cos(2xy) + i \sin(2xy)) \\ &= x \exp(x^2-y^2) \cos(2xy) - y \exp(x^2-y^2) \sin(2xy) \\ &\quad + i [x \exp(x^2-y^2) \sin(2xy) + y \exp(x^2-y^2) \cos(2xy)] \end{aligned}$$

From there:

$$\begin{aligned} U(x,y) = \operatorname{Re} f(z) &= x \exp(x^2-y^2) \cos(2xy) - y \exp(x^2-y^2) \sin(2xy) \\ &= \exp(x^2-y^2) [x \cos(2xy) - y \sin(2xy)] \end{aligned}$$

$$\begin{aligned} V(x,y) = \operatorname{Im} f(z) &= x \exp(x^2-y^2) \sin(2xy) + y \exp(x^2-y^2) \cos(2xy) \\ &= \exp(x^2-y^2) [x \sin(2xy) + y \cos(2xy)] \end{aligned}$$

By taking derivative it is analytic

$$\int_C z \exp(z^2) dz = \int_1^i z \exp(z^2) dz$$

$$= \frac{1}{2} \int_1^1 \exp(w) dw$$

$$\begin{aligned} w &= z^2 \\ dw &= 2z dz \\ 1 \rightarrow i &\rightarrow 1 \rightarrow -1 \end{aligned}$$

Let $f(z) = \sec^2 z$ and $F(z) = \tan z$, Then

Now from ①

$$F'(z) = \sec^2 z = f(z)$$

$$F'(z) = \sec^2 z$$

Now apply integral

$$\begin{aligned} \int \sec^2 z &= \int F'(z) \\ &= F(z) \Big|_{\pi/4}^{\pi/4 i} \\ &= \tan(z) \Big|_{\pi/4}^{\pi/4 i} \\ &= \tan\left(\frac{\pi i}{4}\right) - \tan\left(\frac{\pi}{4}\right) \quad \text{--- ②} \end{aligned}$$

First of all-

$$\tan \frac{\pi}{4} = 1$$

and

$$\begin{aligned} \tan \frac{\pi i}{4} &= \frac{\sinh \frac{\pi i}{4}}{\cosh \frac{\pi i}{4}} \\ &= \frac{e^{i\pi/4} - e^{-i\pi/4}}{e^{i\pi/4} + e^{-i\pi/4}} \\ &= \frac{2i}{2} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{i} \frac{e^{-i\pi/4} - e^{i\pi/4}}{e^{i\pi/4} + e^{-i\pi/4}} \\ &= \frac{1}{i} \times \frac{e^{-i\pi/4} - e^{i\pi/4}}{e^{i\pi/4} + e^{-i\pi/4}} \\ &= \frac{i}{i^2} \frac{-2i \sin \frac{\pi}{4}}{2 \cosh \frac{\pi i}{4}} \\ &= -i \frac{-\sin \frac{\pi}{4}}{\cosh \frac{\pi i}{4}} \\ &= i \tanh \frac{\pi i}{4} \end{aligned}$$

Now put in ②

$$\boxed{= i \tanh \frac{\pi i}{4} - 1}$$

Q 26 $\int (z+z^{-1}) dz$, C the unit circle clockwise.

Solution:-

As we know that for unit circle

$$\begin{aligned} x(t) &= \cos t \rightarrow x(t) = \cos(-t) \\ y(t) &= -\sin t \rightarrow y(t) = \sin(-t) \end{aligned}$$

$$\begin{aligned} z(t) &= \cos t - i \sin t \rightarrow z(t) = \cos(-t) + i \sin(-t) \\ &= e^{-it} \end{aligned}$$

Now putting values

$$\int_0^1 (2(t)) \cdot 2'(t) dt$$

$$= \int_0^1 \operatorname{Re}(z)^2 \cdot z'(t) dz$$

$$= \int_0^1 t^2 \cdot i$$

$$= -i \int_0^1 t^2$$

$$= -i \left[\frac{t^3}{3} \right]_0^1$$

$$= -i \frac{1}{3}$$

$$= -\frac{i}{3}$$

Now

$$= C_1 + C_2 + C_3 + C_4$$

$$= \frac{1}{3} + i - \frac{1}{3} + \frac{1}{3} - 1 - \frac{i}{3}$$

$$= \frac{1 + 3i - 1 + 1 - 3 - i}{3}$$

$$= \frac{-2 + 2i}{3}$$

Q 28 $\int_C \frac{5}{z-2i} - \frac{6}{(z-2i)^2} dz$, C the circle $|z-2i| = 4$ clockwise

Solution

We know that

$$\int (z-z_0)^m dz = \begin{cases} 2\pi i & (m=-1) \\ 0 & (m \neq -1 \text{ and int}) \end{cases}$$

Here $z_0 = 2i$

$$\int_C \frac{5}{z-2i} dz = -5(2\pi i) = -10\pi i$$

$$\int_C \frac{6}{(z-2i)^2} dz = 0$$

Now $\int_C \left(\frac{5}{z-2i} - \frac{6}{(z-2i)^2} \right) dz = -10\pi i$

Q 27 $\int_C \sec^2 z dz$, any path from $\pi/4$ to $\pi/4$

Solution:

$$\frac{d}{dz} \tan z = \sec^2 z \quad \text{--- (1)}$$

$$\int_1^0 t^2 dt$$

$$= \frac{t^3}{3}$$

$$= \frac{1}{3}$$

Along path C_2 :

$$x(t) = (1-t)(1) + t(1) = 1-t+t = 1$$

$$y(t) = (1-t)(0) + t(i) = t$$

$$z(t) = 1 + ti \quad z'(t) = i$$

Now putting values

$$= \int f(z(t)) \cdot z'(t) dz$$

$$= \int \operatorname{Re}(z(t))^2 \cdot z'(t) dz$$

$$= \int_1^0 (1-t^2) \cdot i dt$$

$$= i \int_1^0 1-t^2$$

$$= i \left(\left[t - \frac{t^3}{3} \right]_1^0 \right)$$

$$= i \left(1 - \frac{1}{3} \right) = i - \frac{i}{3}$$

Along C_3 path:

$$x(t) = (1-t)(0) + t(1) = t$$

$$y(t) = (1-t)(1) + t(1) = 1-t+t = 1$$

$$z(t) = t + i \quad z'(t) = 1$$

$$\int f(z(t)) \cdot z'(t) dz$$

$$= \int \operatorname{Re}(z(t))^2 \cdot z'(t) dt$$

$$= \int_0^1 (t+i)^2 \cdot 1 dt$$

$$= \int_0^1 (t^2 + 2ti + i^2) dt$$

$$= \left[\frac{t^3}{3} + 2ti + i^2 t \right]_0^1$$

$$= \frac{1}{3} - 1$$

Along C_4 path

$$x(t) = (1-t)(0) + t(0) = 0$$

$$y(t) = (1-t)(0) + t(1) = t$$

$$z(t) = ti$$

$$z'(t) = i$$

$$2(1-i) \left[\frac{t^3}{3} - \frac{t^2}{2} \right] \Big|_0^1$$

$$= \frac{-1+i}{3}$$

Along C_3 path

$$x(t) = (1-t)(0) + t(0) = 0$$

$$y(t) = (1-t)(1) + t(0) = 1-t$$

$$z(t) = (1-t)i \quad z'(t) = -i$$

Now putting values

$$= \int f(z(t)) z'(t)$$

$$= \int_0^1 i(1-t)^2 \cdot (-i)$$

$$= 0$$

Now C_1 and C_3 is zero but C_2 is answer

$$= \frac{-1+i}{3}$$

Q30 $\int \operatorname{Re} z^2 dz$ clockwise around the boundary of square $0, i, 1+i, 1$

Solution

$$f(z) = \operatorname{Re} z^2 = \operatorname{Re}(u+iy)^2 = \operatorname{Re}(u^2 - y^2 + 2uyi) = u^2 - y^2$$

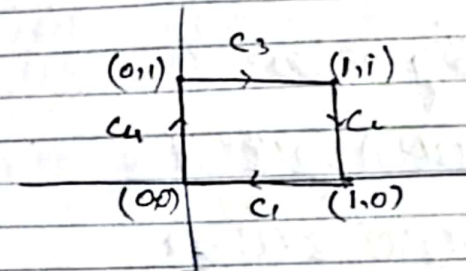
$$V(u, y) = u^2 - y^2$$

$$\frac{\partial V}{\partial u} = 2u$$

$$V(u, y) = 0$$

$$\frac{\partial V}{\partial y} = 0$$

$$\partial u \neq 0$$



Along C_1 path

$$x(t) = (1-t)(0) + t(1) = t$$

$$y(t) = (1-t)(0) + t(0) = 0$$

$$z(t) = t \quad z'(t) = 1$$

Now putting values

$$= \int f(z(t)) \cdot z'(t) dz$$

$$= \int \operatorname{Re}(z(t))^2 \cdot z'(t) dz$$

Q29 $\int \text{Im } z^2 dz$ counterclockwise around the triangle with vertices $0, 1, i$

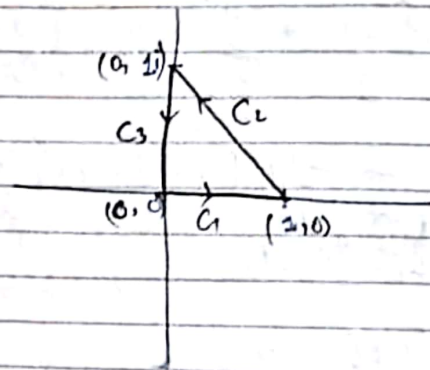
Solution

$$\begin{aligned} f(z) &= f(x+yi) = \text{Im}((x+yi)^2) \\ &= \text{Im}(x^2 - y^2 + 2xyi) \\ &= 2xy \end{aligned}$$

$$U(x,y) = 2xy \quad V(x,y) = 0$$

$$\begin{aligned} \frac{\partial U}{\partial x} &= 2y & \frac{\partial V}{\partial x} &= 0 \\ \frac{\partial U}{\partial y} &= 2x & \frac{\partial V}{\partial y} &= 0 \end{aligned}$$

$$2y \neq 0$$



Along C_1 path:

$$\begin{aligned} x(t) &= (1-t)(0) + t(1) & y(t) &= (1-t)(0) + t(0) \\ &= t & y(t) &= 0 \end{aligned}$$

$$z_1(t) = t \quad z_1'(t) = 1$$

Now putting values

$$\begin{aligned} &= \int f(z_1(t)) z_1'(t) dt \\ &= \int f(t) \cdot 1 dt \\ &= \int \text{Im}(t^2) dt \end{aligned}$$

Along C_2 path:

$$\begin{aligned} x(t) &= (1-t)(1) + t(0) = 1-t \\ y(t) &= (1-t)(0) + t(1) = t \end{aligned}$$

$$\begin{aligned} z(t) &= x(t) + y(t)i \\ &= 1-t + ti \end{aligned}$$

$$z'(t) = -1 + i$$

Now putting values

$$\begin{aligned} &= \int f(z_2(t)) z_2'(t) dt \\ &= \int f(1-t+ti) \cdot (-1+i) dt \\ &= \int \text{Im}((1-t+ti)^2) \cdot (-1+i) dt \\ &= \int \text{Im}((1-t)^2 + (ti)^2 + 2t(1-t)i) \cdot (-1+i) dt \\ &= \int (2t(1-t)) \cdot (-1+i) dt \end{aligned}$$

Q22 $\int \operatorname{Re} z \, dz$, C the parabola $y = 1 + \frac{1}{2}x^2$
from $1+i$ to $3+3i$

Solution

$$f(z) = f(u+yi) = R(u+yi) = u$$

$$\begin{aligned} U(x,y) &= x & V(x,y) &= 0 \\ \frac{\partial U}{\partial x} &= 1 & \frac{\partial V}{\partial x} &= 0 \\ & & & 1 \neq 0 \end{aligned}$$

So f is not analytic first method is not applied

→ Explicitly we have

$$\begin{aligned} x &= t \\ y &= 1 + \frac{1}{2}(t-1)^2 \end{aligned}$$

$$z(t) = t + \left(1 + \frac{1}{2}(t-1)^2\right)i$$

$$z'(t) = 1 + (t-1)i$$

$$\begin{aligned} \int_C f(z) \, dz &= \int_0^3 f(z(t)) \cdot z'(t) \, dt \\ &= \int_0^3 \left(t + \left(1 + \frac{1}{2}(t-1)^2\right)i\right) \cdot (1 + (t-1)i) \, dt \\ &= \int_0^3 \operatorname{Re}\left(t + \left(1 + \frac{1}{2}(t-1)^2\right)i\right) \cdot (1 + (t-1)i) \, dt \\ &= \int_0^3 t(1 + (t-1)i) \, dt \\ &= \int_0^3 (t + it^2 - it) \, dt \\ &= \int_0^3 t \, dt + i \int_0^3 t^2 \, dt - i \int_0^3 t \, dt \\ &= \left[\frac{t^2}{2}\right]_0^3 + i \left[\frac{t^3}{3}\right]_0^3 - i \left[\frac{t^2}{2}\right]_0^3 \\ &= \frac{3^2 - 0}{2} + i \left(\frac{3^3 - 0}{3}\right) - i \left(\frac{3^2 - 0}{2}\right) \\ &= \frac{9}{2} - \frac{1}{2} + i \left(9 - \frac{1}{3}\right) - i \left(\frac{9}{2} - \frac{1}{2}\right) \\ &= 4 + i \cdot \frac{26}{3} - 4i \\ &= 4 + i \cdot \frac{16}{3} \end{aligned}$$

Q 21 $\int \operatorname{Re} z \, dz$, C the shortest path from $1+i$ to $3+3i$

Solution: $f(z) = f(x+iy) = \operatorname{Re}(x+iy) = x$

$$U(x,y) = x \quad V(x,y) = 0$$

$$\frac{\partial U}{\partial x} = 1$$

$$\frac{\partial V}{\partial y} = 0$$

$$1 \neq 0$$

So f is not analytic means we cannot use first method

$$x(t) = (1-t)x_1 + tx_2$$

$$y(t) = (1-t)y_1 + ty_2$$

Now put $x_1=1$ $y_1=1$ $x_2=3$ $y_2=3$

$$x(t) = (1-t)1 + 3t = 1+2t$$

$$y(t) = (1-t)1 + 3t = 1+2t$$

$$z(t) = (1+2t) + (1+2t)i$$

$$\bar{z}(t) = 2+2i$$

$$\int_C f(z) \, dz = \int_0^1 f(z) \bar{z}(t) \, dt$$

$$= \int_0^1 f(1+2t + (1+2t)i) \cdot (2+2i) \, dt$$

$$= (2+2i) \int_0^1 \operatorname{Re}(1+2t + (1+2t)i) \, dt$$

$$= 2+2i \int_0^1 (1+2t) \, dt \quad \text{--- (1)}$$

$$\int_0^1 (1+2t) \, dt = \int_0^1 1 \, dt + \int_0^1 2t \, dt$$

$$= t \Big|_0^1 + t^2 \Big|_0^1$$

$$\int_0^1 (1+2t) \, dt = 1 + 1 = 2$$

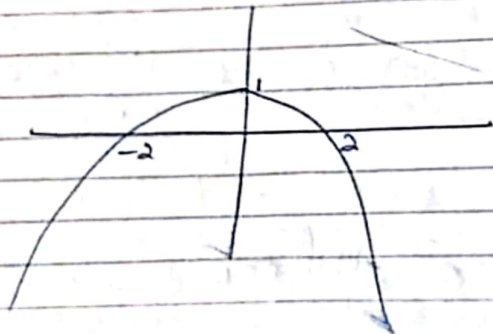
Now put this in (1)

$$= 2(2+2i)$$

$$= 4+4i$$

For ending point $(-2, -2)$

$$\begin{aligned}x(-2) &= 2 \\y(-2) &= 1 - \frac{(-2)^2}{4} \\&= 1 - \frac{4}{4} = 0\end{aligned}$$



Q20 $4(x-2)^2 + 5(y+1)^2 = 20$

Solution

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$x(t) = h + a \cos(t)$$

$$y(t) = k + b \sin(t)$$

$$\frac{(x-2)^2}{5} + \frac{(y+1)^2}{4} = 1$$

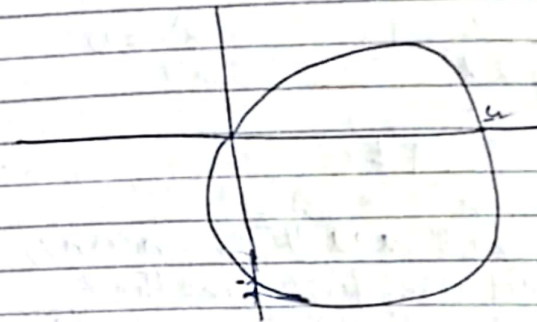
$$\frac{(x-2)^2}{\sqrt{5}} + \frac{(y+1)^2}{2^2} = 1$$

$$a = \sqrt{5} \quad b = 2 \quad (h, k) = (2, -1)$$

$$x(t) = 2 + \sqrt{5} \cos t$$

$$y(t) = -1 + 2 \sin t$$

$$z(t) = (2 + \sqrt{5} \cos t) + i(-1 + 2 \sin t)$$



Q18 $y = \frac{1}{x}$ from $(1,1)$ to $(5, \frac{1}{5})$

Solution

Explicitly written as function

$$x(t) = t$$

Then

$$y(t) = \frac{1}{t}$$

$$z(t) = t + \frac{i}{t}$$

Now Starting point $(1,1)$

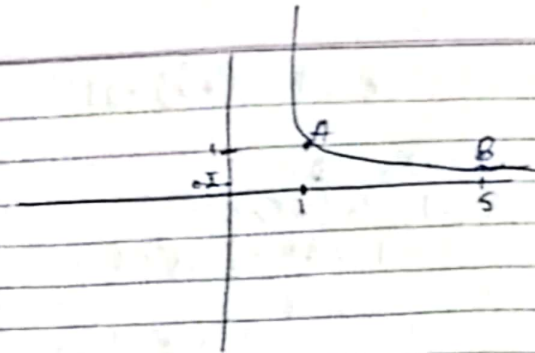
$$x(1) = 1$$

$$y(1) = \frac{1}{1} = 1 \quad (1,1)$$

Now ending point

$$x(5) = 5$$

$$y(5) = \frac{1}{5} \quad (5, \frac{1}{5})$$



Q19 Parabola $y = 1 - \frac{1}{4}x^2$ $(-2 \leq x \leq 2)$

Solution:

$$x = t$$

$$y = 1 - \frac{t^2}{4}$$

$$z(t) = t + i\left(1 - \frac{t^2}{4}\right)$$

For Starting point $(-2, -2)$

$$x = -2$$

$$y = 1 - \frac{(-2)^2}{4}$$

$$= 1 - \frac{4}{4}$$

$$= 1 - 1 = 0$$

$$\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$$

$$a=3 \quad b=2 \quad (h,k)=(0,0)$$

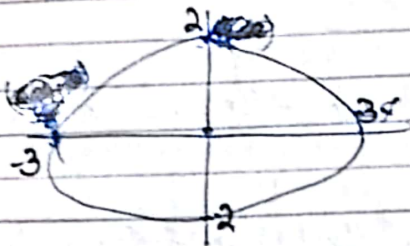
Now

$$x(t) = 0 + 3 \cos t = 3 \cos t$$

$$y(t) = 0 + 2 \sin t = 2 \sin t$$

Here

$$z(t) = 3 \cos t + i 2 \sin t$$



Q 17 $|z+a+ib|=r$ clockwise

Solution

$$|z+a+ib|=r$$

Let $z=x+iy$ for real number x and y
Then

$$|z+a+ib| = |(x+a) + (y+b)i| = \sqrt{(x+a)^2 + (y+b)^2}$$

$$\sqrt{(x+a)^2 + (y+b)^2} = r$$

$$(x+a)^2 + (y+b)^2 = r^2$$

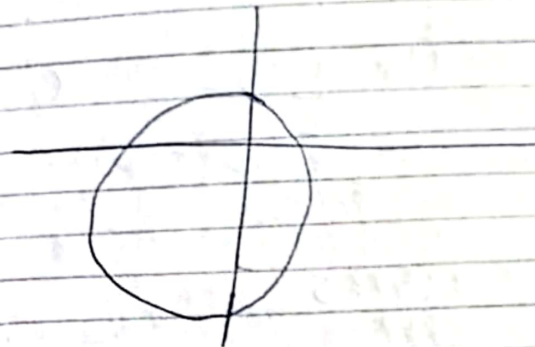
Now from this eqn

$$x(t) = -a + r \cos t$$

$$y(t) = -b + r \sin t$$

$$z(t) = (-a + r \cos t) + i(-b + r \sin t)$$

Now we are sketching with
 $a=1, b=2, r=1$



Q14 Unit circle clockwise

Solution

$$\begin{aligned}x(t) &= r \cos t \\ y(t) &= r \sin t\end{aligned}$$

but for unit circle $[r=1]$ so

$$\begin{aligned}x(t) &= \cos t \\ y(t) &= \sin t\end{aligned}$$

Put these

$$\begin{aligned}z(t) &= x(t) + iy(t) \\ z(t) &= \cos t + i \sin t\end{aligned}$$



Q15

$x^2 + 4y^2 = 4$ the branch through $(2,0)$

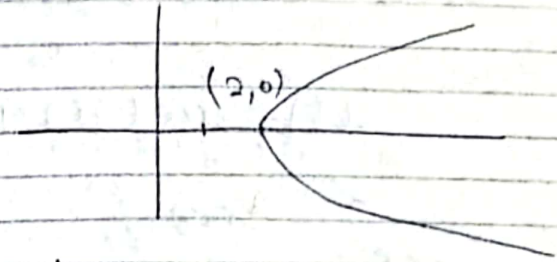
Solution

$$\frac{x^2}{4} + \frac{y^2}{1} = 1 \Rightarrow \frac{x^2}{4} + \frac{y^2}{1} = 1$$

We can parameterize that

$$\begin{aligned}x(t) &= a \cos t \Rightarrow x(t) = 2 \cos t \\ y(t) &= b \sin t \Rightarrow y(t) = \sin t\end{aligned}$$

$$z(t) = 2 \cos t + i \sin t$$



Q16 Ellipse $4x^2 + 9y^2 = 36$ counterclockwise

Solution:

We know that ellipse given in form

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

and we can parameterize

$$\begin{aligned}x(t) &= h + a \cos(t) \\ y(t) &= k + b \sin(t)\end{aligned}$$

(h,k) is centre of ellipse
 a is radius along x
 b is radius along y

Now

$$\begin{aligned}4x^2 + 9y^2 &= 36 \\ \frac{x^2}{9} + \frac{y^2}{4} &= 1\end{aligned}$$

Parametric Representation
and sketch the path

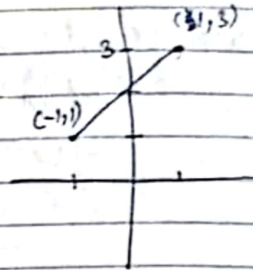
Q11 Segment from $P(-1,1)$ to $(1,3)$

Sol

$$V = (1+1, 3-1)$$

$$V = (2, 2)$$

$$P = (-1, 1)$$

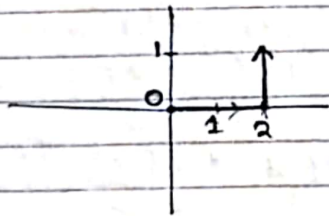


$$\begin{aligned} z(t) &= P + Vt \\ &= (-1, 1) + (2, 2)t \\ &= (-1, 1) + (2t, 2t) \\ &= (-1+2t) + i(1+2t) \end{aligned}$$

Q12 From $(0,0)$ to $(2,1)$ along the axes

$$x(t) = (0, 2)$$

$$y(t) = (0, 1)$$



Q13 Upper half of $|z-2+i|=2$ from $(4,-1)$ to $(0,1)$

Sol

$$|z-2+i|=2$$

Write $z = x+yi$ for some real numbers

$$|z-2+i| = |(x-2) + (y+1)i| = \sqrt{(x-2)^2 + (y+1)^2}$$

So we have

$$|(x-2) + (y+1)i| = |z-2+i| = 2$$

$$\sqrt{(x-2)^2 + (y+1)^2} = 2$$

$$(x-2)^2 + (y+1)^2 = 2^2$$

$$(x-2)^2 + (y+1)^2 = 4$$

From this eqn

$$x(t) = 2 + 2 \cos t$$

$$y(t) = -1 + 2 \sin t$$

$$z(t) = x + yt$$

$$z(t) = (2 + 2 \cos t) + i(-1 + 2 \sin t)$$

